

- 5 (a) Fig. 5.1 is a photomicrograph of a section through the lungs showing a bronchus and some alveoli.

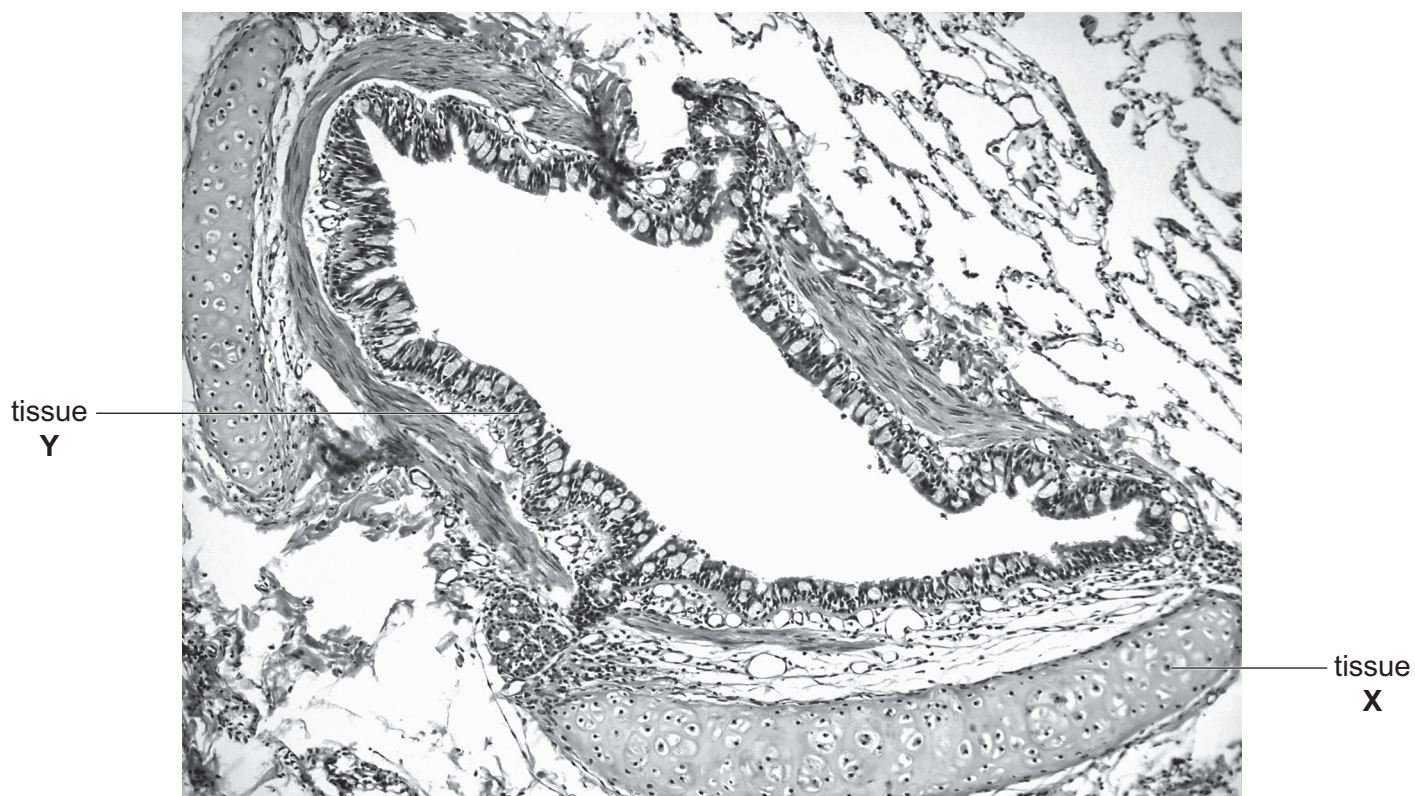


Fig. 5.1

- (i) State the function of tissue **X** labelled in Fig. 5.1.
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- [1]
- (ii) Describe how the distribution of tissue **X** in the trachea differs from that shown in Fig. 5.1.
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- [1]
- (iii) Describe how the epithelial tissue, **Y**, is adapted for its function.
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- [2]

(b) Fig. 5.2 is a photograph of two African elephants, *Loxodonta africana*.



Fig. 5.2

- (i) Describe the difference in surface area to volume ratio between the adult elephant and baby elephant shown in Fig. 5.2.

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..... [1]

- (ii) Suggest why animals such as elephants require a gas exchange system.

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..... [3]

- (iii) The feet of elephants are protected by structures under the skin known as cushions. The cushions are made up of a large number of cells surrounded by connective tissue containing many fibres of collagen. The fibres help to maintain the structure of the cushion.

The collagen fibres are made of collagen molecules.

Describe the structure of a **collagen molecule**.

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- (iv) The cushion in the foot is very strong and is able to resist extremely large forces acting on it due to the large mass of the elephant.

Suggest how the structure of a **collagen fibre** can help the cushion resist these large forces.

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[Total: 13]